

Bayesian Hill Mixture Models for Marketing Mix Modeling: Synthetic Benchmarks, Component Resolvability, and Real-Data Evaluation

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Abstract

Marketing Mix Models (MMMs) typically assume that all consumers respond to advertising with a single Hill saturation curve, but real audiences are heterogeneous. We study a Bayesian mixture of K Hill curves implemented in NumPyro, with an ordering reparameterisation, post-hoc relabeling, and label-invariant convergence diagnostics designed to make mixture posteriors interpretable.

We report three experiments. (i) A controlled synthetic benchmark shows that mixture models improve expected log pointwise predictive density (ELPD-LOO) whenever the true data-generating process is heterogeneous, and incur only a small penalty otherwise. (ii) A *component resolvability* study varies the cosine distance between true Hill curves and identifies an empirical threshold of $\bar{d}_{\cos} \approx 0.10$, below which posteriors collapse to fewer effective components. (iii) A real-data benchmark on three Conjura organisations across three verticals (beauty & fitness, home & garden, toys & hobbies) shows that Mixture $K=3$ improves ELPD-LOO over single-Hill in every organisation, by 222 to 308 units, with the gain primarily in predictive density rather than point error.

The strict publication-ready convergence criterion that all 45 synthetic fits satisfy fails uniformly for Mixture $K=3$ on real data. Re-using the resolvability axis on the real posteriors shows that this gap is organisation-dependent: Home & Garden sits at the resolvability boundary, Toys & Hobbies carries three genuinely resolvable components but with insufficient chain mixing, and Beauty & Fitness shows strong heterogeneity coupled with influential observations. We report these three regimes directly rather than collapsing them into a single summary. Code, configurations, and per-fit artifacts are released for reproduction.

Keywords: Marketing Mix Modeling, Bayesian inference, Hill function, mixture models, label switching, component resolvability

1 Introduction

Marketing Mix Modeling (MMM) enables organisations to quantify the effectiveness of advertising spend and to optimise budget allocation. The canonical formulation fits a single response curve per channel on aggregate time-series data, combining geometric (or Weibull) adstock for carryover with a Hill function for saturation [Jin et al., 2017, Chan and Perry, 2017]. We refer to this baseline as *Standard MMM*.

Standard MMM assumes that all consumers respond to a given channel with the same saturation curve. In practice, audiences are heterogeneous: heavy versus light buyers, loyalists versus switchers, and geographically distinct segments exhibit qualitatively different response patterns. A

single aggregate curve is a weighted average of these latent responses, and the aggregate estimate can be misleading even when in-sample fit looks good. This paper studies a Bayesian mixture of Hill curves as a principled extension of Standard MMM that makes the latent heterogeneity explicit, and asks two empirical questions: how much heterogeneity must be present before the mixture can recover it, and how well do the recovery guarantees that hold on synthetic data transfer to field data?

Contributions.

1. A controlled synthetic benchmark comparing single-Hill and Bayesian mixture-of-Hill models under known ground truth, with predictive, latent-recovery, and convergence metrics.
2. A *component resolvability* study that systematically varies the cosine distance between true Hill curves and traces the recovered Shannon effective component count as a function of true separation, identifying an empirical threshold below which the posterior collapses to fewer components.
3. A real-data benchmark on three Conjura organisations across three verticals, paired with a re-use of the resolvability axis on the real posteriors. The combination shows that mixture-of-Hill improves predictive density uniformly across organisations but that publication-ready convergence is organisation-dependent in a way the resolvability framework can diagnose.

Roadmap. Section 2 situates the contribution against the MMM, Bayesian segmentation, and mixture-identification literatures. Section 3 specifies the model and the label-invariant diagnostic workflow. Section 4 consolidates the inference configuration, evaluation metrics, and publication-ready convergence criterion used throughout the paper. Sections 5–7 report the three benchmarks. Section 8 discusses what worked and what did not, Section 9 states limitations, and Section 10 describes the released artifacts.

2 Related Work

Bayesian MMM with adstock and saturation. The combination of geometric adstock with Hill saturation under a Bayesian likelihood is the canonical modern MMM formulation [Jin et al., 2017, Chan and Perry, 2017]. Subsequent open-source implementations (PYMC-MARKETING, MERIDIAN) have made this family practical at industrial scale; we keep our model in the same family so that the mixture extension is a minimal, drop-in change rather than a separate architecture. We deliberately do not extend the dynamics: Dew et al. [2024] argue that nonlinear and time-varying effects in MMM are jointly not identified from typical observational time series, so apparent saturation can be an artefact of misspecified dynamics. Their critique motivates our *controlled* treatment of nonlinearity, in which heterogeneity rather than dynamics is the source of curve shape.

Heterogeneous response and segmentation. The marketing literature has long emphasised consumer heterogeneity [Wedel and Kamakura, 2000, Allenby and Rossi, 1998]. Hierarchical Bayes and latent-class models are the standard tools at the individual or panel level. MMM, by contrast, operates on aggregated time series and typically suppresses heterogeneity. Mixture-of-Hill is one of the smallest extensions that brings explicit latent segments back into the aggregate model without leaving the standard MMM family.

Mixture identifiability and label switching. Finite-mixture posteriors are non-identified under permutation of component labels [Stephens, 2000, Frühwirth-Schnatter, 2006]. The label-switching problem makes per-component summaries meaningless without an additional convention, and breaks standard split- $\hat{R}$ diagnostics [Vehtari et al., 2021]. The two standard remedies—ordering constraints and post-hoc relabeling [Papastamoulis, 2014]—trade off detailed balance against interpretability. We combine both with an additional label-invariant diagnostic (Section 3.1) and treat them as complementary rather than substitutable.

3 Model Specification

Let x_t denote aggregated marketing spend at time t and y_t the observed outcome (purchases, revenue, or a related target). Figure 1 shows the model in plate notation. The model is implemented in NumPyro [Phan et al., 2019] with JAX acceleration.

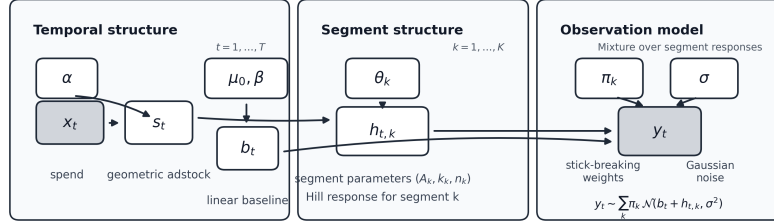


Figure 1: Graphical model. Observed variables are shaded; plates indicate replication over time steps t and mixture components k .

Adstock transformation. Geometric decay captures carryover from previous periods:

$$s_t = x_t + \alpha \cdot s_{t-1}, \quad s_0 = 0, \quad \alpha \sim \text{Beta}(2, 2). \quad (1)$$

Hill saturation. Each mixture component $k \in \{1, \dots, K\}$ has its own Hill response:

$$f_k(s) = A_k \cdot \frac{s^{n_k}}{\lambda_k^{n_k} + s^{n_k}}, \quad (2)$$

with maximum effect A_k , half-saturation point λ_k , and slope $n_k \sim \text{LogNormal}(\log 1.5, 0.4)$.

Mixture likelihood. The observation model is a Gaussian mixture over components:

$$y_t \sim \sum_{k=1}^K \pi_k \cdot \mathcal{N}(\mu_0 + \beta t + f_k(s_t), \sigma^2), \quad (3)$$

where $\pi \sim \text{Dirichlet}(\mathbf{1}_K)$ and μ_0, β capture intercept and trend.

Automatic prior scaling. To accommodate diverse data scales, priors on A_k , μ_0 , and σ are computed from training-data statistics: $A_k \sim \text{LogNormal}(\log(0.3 \cdot \text{range}(y)), 0.8)$, $\mu_0 \sim \mathcal{N}(\bar{y}, 2\sigma_y)$, $\sigma \sim \text{HalfNormal}(\sigma_y)$.

3.1 Identifiability and Label Switching

A finite mixture is invariant under permutations of component labels: any reordering of $(\pi_k, A_k, \lambda_k, n_k)$ yields the same likelihood. As a result, posterior summaries computed per component are meaningless without an additional convention, and standard split- $\hat{R}$ on raw component parameters cannot distinguish genuine non-mixing from label permutations across chains [Stephens, 2000, Frühwirth-Schnatter, 2006].

Within-MCMC ordering. We impose $\lambda_1 < \lambda_2 < \dots < \lambda_K$ via cumulative-sum reparameterization with positive increments $\delta_j \sim \text{LogNormal}(\log(s_{\max}/(K+1)), 0.7)$. This restricts the posterior to one canonical labelling but can distort posterior geometry near the boundary.

Post-hoc relabeling. We additionally implement unconstrained sampling followed by post-hoc reordering: each posterior draw is permuted so that $\lambda_1 < \lambda_2 < \dots < \lambda_K$ [Stephens, 2000, Papastamoulis, 2014]. This preserves detailed balance and exposes interpretable per-component summaries.

Label-invariant diagnostics. We compute $\hat{R}$ and effective sample sizes on three groups of quantities [Vehtari et al., 2021]:

1. *Joint log-likelihood* $\sum_t \log \sum_k \pi_k \phi(y_t \mid \mu_t^{(k)}, \sigma)$, which is label-invariant by construction.
2. *Scalar parameters* $(\alpha, \sigma, \mu_0, \beta)$ that are not affected by label permutations.
3. *Relabeled component parameters*, after the post-hoc reordering above.

A model is treated as effectively converged when all three pass; this is the criterion we use to report a *publication-ready* fit. Concretely, a fit is publication-ready when (i) the No-U-Turn sampler produces zero divergences, zero maximum-tree-depth saturations, and a minimum chain-level BFMI of at least 0.30; and (ii) every retained mixing diagnostic satisfies $\hat{R} \leq 1.01$ with both bulk and tail effective sample size of at least 100 per chain. The mixing diagnostics retained depend on the model: for single-Hill we use raw parameter $\hat{R}/\text{ESS}$; for mixtures we replace the raw component-parameter diagnostics with the label-invariant diagnostics above and additionally require the relabeled component parameters to satisfy the same thresholds. These thresholds match the conservative defaults of Vehtari et al. [2021] and ensure that “pass” is a strict criterion rather than a soft one.

4 Experimental Setup

Inference. All fits use the No-U-Turn Sampler (NUTS) with two chains, init strategy `uniform`, and dense mass adaptation. Warmup and sampling lengths and the target acceptance probability are model-dependent; on synthetic data we use (600, 600) for single-Hill on the $K = 1$ and $K = 2$ DGPs and (800, 800) on the $K = 3$ DGP, (900, 900) for Mixture $K = 2$, and (1,600, 1,200) for Mixture $K = 3$. On real data the same per-model schedule is used with (600, 600), (900, 900), and (1,600, 1,200) respectively. Target acceptance is increased for larger mixtures to control divergences. The complete configuration is reproduced in Appendix A.

Data splits. Each time series is divided into a 75/25 train/test split. Holdout predictions inherit the adstock state accumulated during training and are evaluated at their absolute time index rather than reset to zero, so that adstock decay and the trend term are propagated forward consistently.

Evaluation metrics. We report (i) expected log pointwise predictive density estimated by Pareto-smoothed importance-sampling leave-one-out cross-validation (ELPD-LOO), with the count of $\hat{k}_i > 0.7$ observations reported alongside as a reliability flag; (ii) test-set mean absolute percentage error (MAPE), continuous ranked probability score (CRPS), and 90% predictive coverage; and (iii) for synthetic conditions, latent-mean normalised RMSE against the known DGP latent.

Publication-ready convergence. The criterion defined in Section 3.1 is applied identically across all experiments: zero divergences, zero max-tree-depth saturations, $\text{BFMI} \geq 0.30$ on every chain, $\hat{R} \leq 1.01$, and bulk/tail ESS per chain ≥ 100 on the retained mixing diagnostics (raw parameters for single-Hill; label-invariant joint log-likelihood plus scalar non-component parameters, together with relabeled component parameters, for mixtures).

Component summaries. For mixture fits we additionally report posterior mixture weights $\hat{\pi}_k$, the Shannon effective component count ${}^1D = \exp(-\sum_k \hat{\pi}_k \log \hat{\pi}_k)$ [Hill, 1973], and the mean pairwise cosine distance between active Hill response curves. The latter is computed on a fit-adaptive saturation grid that covers the active components’ half-saturation range, making it independent of how the prior scale was set.

5 Synthetic Benchmark

5.1 Data-Generating Processes

We evaluate the models across three data-generating processes (DGPs): (1) *Single* ($K = 1$), a single Hill response; (2) *Mixture* $K = 2$, a two-component mixture; and (3) *Mixture* $K = 3$, a three-component mixture. Each DGP uses $T = 200$ observations and we run five random seeds per condition, so the design contains $3 \times 3 \times 5 = 45$ fits.

5.2 Results

Convergence. Under the benchmark threshold $\hat{R} \leq 1.05$ on raw parameters for single-Hill and on label-invariant diagnostics for mixtures, 44 of 45 synthetic fits cleared the threshold (Figure 2). The single exception was the Mixture $K = 3$ model on the $K = 2$ DGP at seed 1, which reported a max $\hat{R} \approx 1.5$ and is flagged as a benchmark failure rather than silently included. Standard single-Hill diagnostics agreed with the mixture-specific label-invariant diagnostics on every other synthetic fit. The stricter publication-ready criterion ($\hat{R} \leq 1.01$, ESS ≥ 100 per chain, plus the HMC clauses of Section 3.1) is more demanding: 33 of 45 fits cleared it, with the remaining cases recorded as warnings (typically $\hat{R}$ in the range $[1.01, 1.05]$). We retain the strict criterion as the headline figure for real data in Section 7, but the synthetic Section 5 plots use the looser benchmark threshold so that the comparison against published MMM benchmarks is on equal footing.

Predictive performance. Figures 3 and 4 report ELPD-LOO and holdout CRPS. When the true DGP is homogeneous ($K = 1$), the three models tie within standard error. When heterogeneity is present, mixtures gain substantially in ELPD ($\Delta \approx +50$ on $K = 2$, $\Delta \approx +45$ on $K = 3$). CRPS

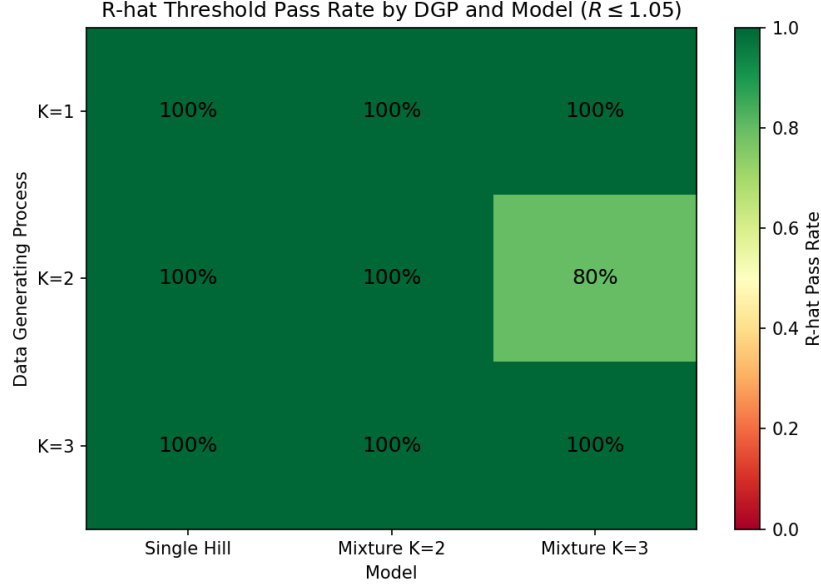


Figure 2: Benchmark convergence pass rate ($\hat{R} \leq 1.05$ on the retained mixing diagnostics) by DGP and model on the synthetic benchmark. Eight cells reach 100%; the Mixture $K = 3$ model on the $K = 2$ DGP reaches 80% (4 of 5 seeds), with seed 1 failing at max $\hat{R} \approx 1.5$.

gains are noticeably smaller than ELPD gains ($\Delta \approx 0.1$ on $K = 2$, $\Delta \approx 0.4$ on $K = 3$), indicating that the bulk of the improvement is in probabilistic density rather than in point prediction.

Latent recovery. Figure 5 shows latent-mean nRMSE on the test split. The pattern mirrors ELPD: mixtures roughly halve nRMSE under $K = 2$ ($0.14 \rightarrow 0.07$) and reduce it by a factor of ≈ 2.4 under $K = 3$ when the model is matched to the true K ($0.085 \rightarrow 0.035$). This shows that the ELPD gains correspond to genuinely better recovery of the latent response, not just better marginal calibration.

6 Component Resolvability

The previous section shows that mixtures help when heterogeneity exists, but does not tell us *how much* heterogeneity is required for the posterior to actually separate components. We address this with a controlled study.

6.1 Design

We construct nine DGP profiles by varying the spacing δ between true Hill half-points around a fixed centre c , holding amplitude and slope fixed. For $K_{\text{true}} = 1$ we use one anchor profile; for $K_{\text{true}} \in \{2, 3\}$ we use four profiles each, spanning low to extreme separation. Each profile is run with five seeds and fit with both Mixture $K = 2$ and Mixture $K = 3$, for $9 \times 2 \times 5 = 90$ fits in total. The separation axis $\bar{d}_{\text{cos}}$ and recovery axis 1D are the cosine-separation and Shannon-effective-count measures defined in Section 4; here $\bar{d}_{\text{cos}}$ is computed on the *true* component curves and 1D on the posterior mixture weights.

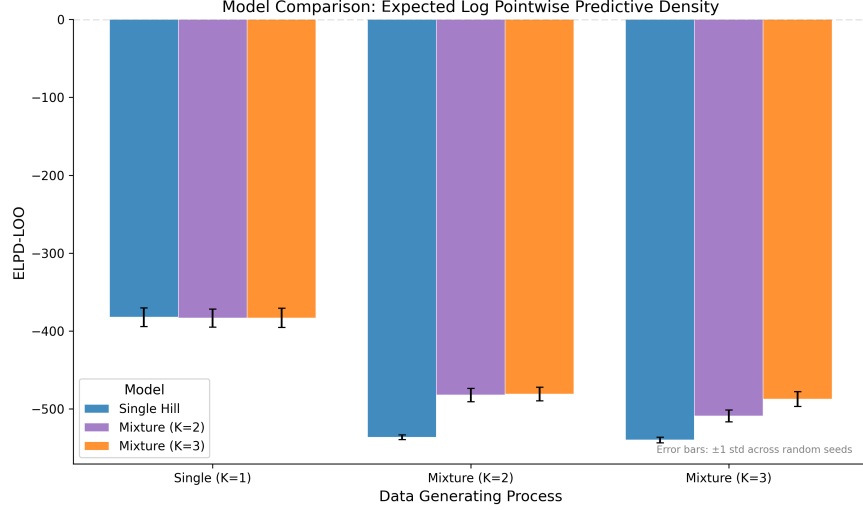


Figure 3: ELPD-LOO by DGP and model on the synthetic benchmark. Higher is better. Mixtures improve substantially when $K \geq 2$.

6.2 Results

Figure 6 plots 1D against $\bar{d}_{\cos}$ across all 90 fits. Three regimes are visible:

- For $\bar{d}_{\cos} \lesssim 0.05$, both models collapse toward the trivial value: even when $K_{\text{true}} \geq 2$, the posterior cannot resolve more than ≈ 1.5 – 1.7 effective components.
- Between $\bar{d}_{\cos} \approx 0.05$ and 0.3 , the recovered count rises sharply with separation: this is the resolvable transition region.
- For $\bar{d}_{\cos} \gtrsim 0.3$, the recovered count saturates near K_{true} . Mixture $K = 2$ saturates near 2 by construction; Mixture $K = 3$ tracks the true K and approaches 3 on $K = 3$ DGPs.

Failure modes. The publication-ready convergence criterion fails most frequently in two regimes. First, under-specified fits at low separation: Mixture $K = 2$ on $K_{\text{true}} = 3$ DGPs at low δ shows publication-fail rates of 0.6–0.8. Second, at the highest separation under $K_{\text{true}} = 3$, Mixture $K = 3$ exhibits a non-trivial failure rate (0.6 at the most extreme profile), suggesting that the strict diagnostic criterion is sensitive to posterior geometry even when the components are far apart.

Practical implication. Combined with the ELPD results of Section 3, the resolvability picture suggests that mixture-of-Hill models can be used reliably when the cosine separation between true components exceeds ≈ 0.1 . Below this threshold, the posterior is consistent with a smaller effective K , and the modeller should expect both lower convergence pass rates and very limited information gain over a single-Hill baseline.

7 Real-Data Benchmark

7.1 Dataset and Organisation Selection

We use the public Conjura MMM dataset, which contains daily spend across nine advertising channels and several purchase-related target variables for a panel of e-commerce organisations.

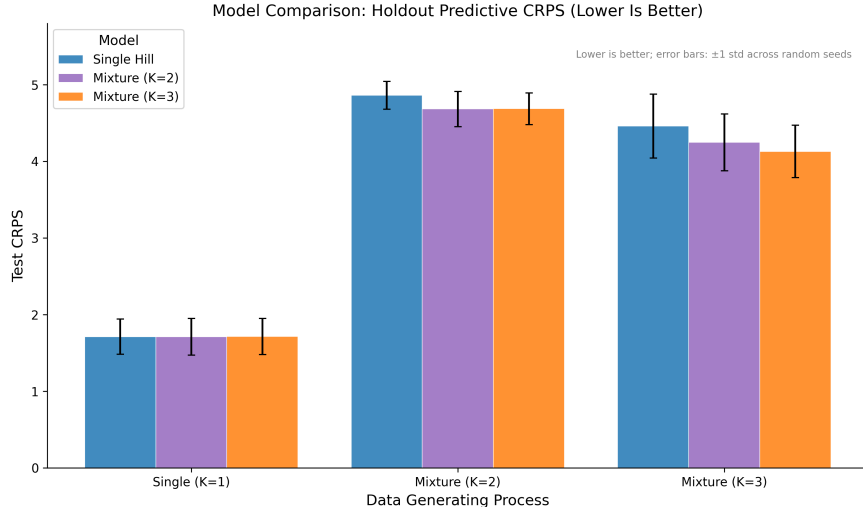


Figure 4: Holdout CRPS by DGP and model on the synthetic benchmark. Lower is better. Gains are smaller in magnitude than ELPD gains.

Following the per-organisation modelling convention of MMM practice, we treat each organisation as an independent time series and do not share parameters across organisations.

To select organisations for the benchmark, we screen all candidates with at least 200 daily observations and at least two active spend channels (non-zero on more than 10% of days), and rank them by a composite score combining series length, spend coefficient of variation, spend–target correlation, and channel breadth. We pick three top-ranked organisations from three different verticals to support a modest cross-vertical claim. The selected organisations are summarised in Table 1.

Table 1: Selected Conjura organisations for the real-data benchmark.

Vertical	T (days)	Active channels	Spend–target r	Total spend (USD M)
Toys & Hobbies	1,375	5	0.95	3.3
Beauty & Fitness	924	8	0.96	5.2
Home & Garden	1,642	5	0.80	2.1

7.2 Predictive Performance

Each organisation is fit with single-Hill, Mixture $K = 2$, and Mixture $K = 3$ under three seeds, giving 27 fits in total; inference and evaluation follow Section 4.

Table 2 reports ELPD-LOO, holdout MAPE, holdout CRPS, and holdout 90% coverage averaged across seeds. The key observation is that *Mixture $K = 3$ achieves the best ELPD-LOO in every organisation*, with improvements of +222 (beauty & fitness), +308 (home & garden), and +275 (toys & hobbies) over the single-Hill baseline. Mixture $K = 2$ is intermediate.

The single-Hill-to-Mixture $K = 3$ ELPD gaps (+222 to +308) are between two and seven LOO standard errors in magnitude, so they are meaningful at fit-level even before averaging across seeds. The PSIS-LOO standard errors are not strictly differences-of-LOOs (`se.diff` from Vehtari et al. [2017]); we report per-fit SE as a conservative lower bound on uncertainty.

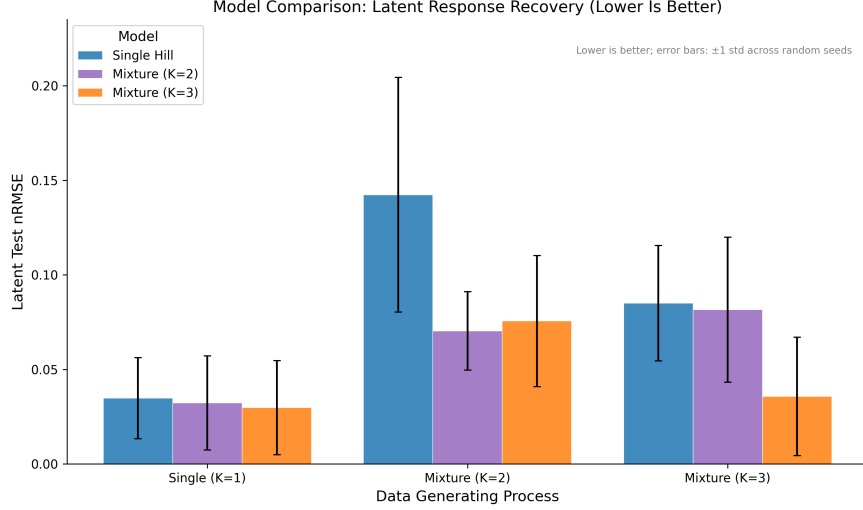


Figure 5: Latent-mean nRMSE by DGP and model. Lower is better. Mixtures recover the latent response more accurately when heterogeneity is present.

Figure 7 visualises the same comparison. The Mixture $K = 3$ advantage is consistent in sign across organisations, while the absolute scale of ELPD-LOO varies by an order of magnitude with series length and target variance.

The relationship between ELPD and the other metrics is heterogeneous. On *home & garden* the rank ordering of models is consistent across ELPD, CRPS, and MAPE: the mixture improves all three. On *toys & hobbies*, ELPD and CRPS both improve, while MAPE is non-monotonic. On *beauty & fitness*, ELPD improves substantially but MAPE and CRPS deteriorate slightly for $K = 3$, and the model remains visibly underconfident: 90% coverage rises from 0.33 (single) to 0.54 ($K = 3$) but stays well below the nominal level, indicating that the predictive intervals are too narrow even for the best fit. We return to this in the discussion.

To make concrete what the mixture is doing, Figure 8 shows the posterior response curves for the home & garden organisation under all three models on a single representative seed. Single-Hill recovers one slowly-saturating curve. Mixture $K = 2$ resolves two components with weights 0.93 and 0.07, in which the small minority component carries a much higher saturation ceiling. Mixture $K = 3$ uses two active components with weights 0.84 and 0.14 and a third component that collapses to weight 0.02 (labelled *inactive* in the legend). The qualitative two-mode structure is robust to whether K is set to 2 or 3, which is consistent with the posterior effective component count reported below.

7.3 Convergence on Real Data

Table 3 summarises the rate at which each (organisation, model) cell passes the publication-ready convergence criterion (HMC diagnostics plus label-invariant and relabeled $\hat{R}$) across the three seeds. Two patterns are stark.

First, *Mixture $K = 3$ passes the publication-ready criterion in 0/9 fits* on real data, despite passing in 100% of the synthetic fits at matched K . Second, the posterior expectation of the Shannon effective count, $\mathbb{E}[^1D]$ (computed per posterior sample and averaged), is at most 2.45, suggesting that draws rarely use the full $K = 3$ capacity. The breakdown is more informative than the aggregate: Section 7.4 below shows that this average hides three qualitatively different

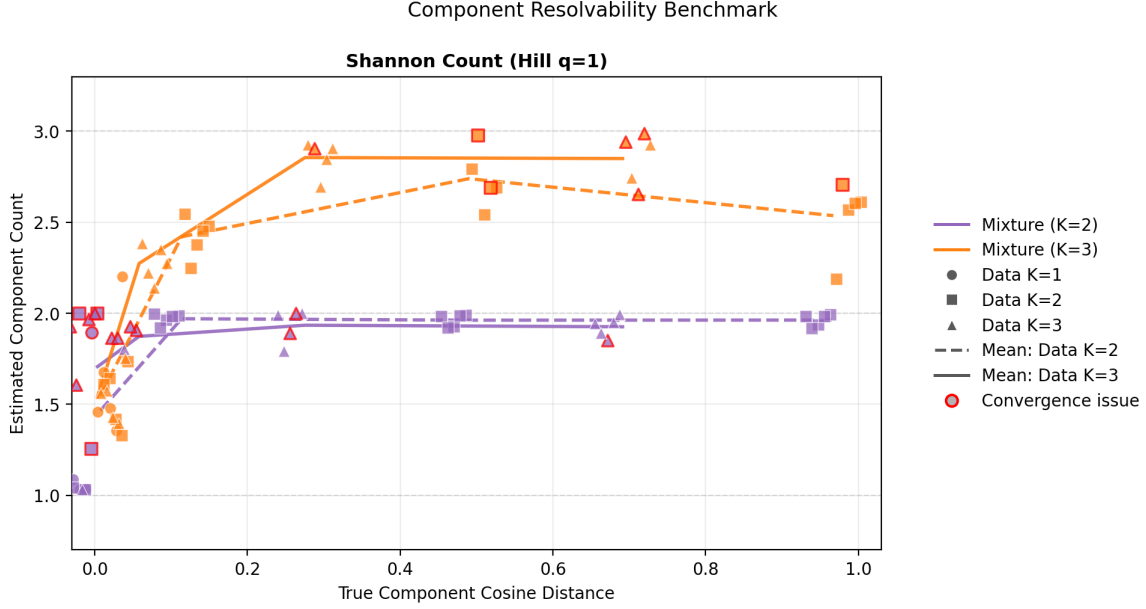


Figure 6: Component resolvability: Shannon effective count of the posterior versus mean pairwise cosine distance of the true Hill curves. Red-outlined points indicate cases that failed the publication-ready convergence criterion.

organisation-level behaviours.

The high Pareto- k count for the beauty & fitness Mixture $K = 2$ fit (15 observations on average) is consistent with a moderately misspecified model on data with influential outliers, and points to robustifying the likelihood as a useful next step.

7.4 Where Real Organisations Sit on the Resolvability Axis

The synthetic resolvability study (Section 6) characterises recovery as a function of *true* cosine separation between component curves. Real data has no ground truth, but we can compute the same quantity from each fit’s *posterior* component summary: the mean pairwise cosine distance between the active Hill curves implied by the posterior means (A_k, λ_k, n_k) , on a fit-adaptive saturation grid. Plotting the recovered Shannon effective count against this posterior separation places the real organisations onto the same plane as the synthetic resolvability fits (Figure 9).

The picture is heterogeneous across organisations. *Home & Garden* sits squarely at the resolvability transition: its posterior cosine separation is 0.09 under Mixture $K = 2$ and 0.12 under Mixture $K = 3$, and its Shannon count stays at 1.27 and 1.66 respectively. This is the regime in which Section 6 predicts collapse to fewer effective components, and is consistent with the effective K near 2 we report in Table 3. *Toys & Hobbies* under Mixture $K = 3$, by contrast, has a posterior separation of 0.29 and a Shannon count of 2.72—a position firmly inside the resolvable region, indicating that the three-component structure is genuinely supported by these data even though the strict convergence criterion still fails. *Beauty & Fitness* occupies the opposite extreme: the highest posterior separation of any organisation (0.46 for Mixture $K = 2$, 0.68 for Mixture $K = 3$), but every fit fails publication-readiness. Read alongside the high Pareto- k counts, this points to genuine multi-modal structure in the response curve coupled with a posterior whose chains do not mix sufficiently to certify that structure.

Table 2: Real-data predictive performance, averaged across three seeds. ELPD-LOO is reported with the per-fit PSIS-LOO standard error in parentheses (averaged across seeds); higher is better. MAPE/CRPS lower is better; 90% coverage closer to 0.9 is better. Bold marks the best ELPD-LOO per organisation.

Organisation	Model	ELPD-LOO ($\pm$ SE)	Test MAPE	Test CRPS	90% cov.
Beauty & Fitness	Single Hill	$-3,008 \pm 86$	30.3	108	0.33
	Mixture $K = 2$	$-2,992 \pm 96$	29.6	101	0.37
	Mixture $K = 3$	$-2,786 \pm 48$	34.3	122	0.54
Home & Garden	Single Hill	$-6,168 \pm 63$	26.0	25.0	0.87
	Mixture $K = 2$	$-5,985 \pm 56$	31.4	27.2	0.88
	Mixture $K = 3$	$-5,860 \pm 48$	28.2	25.9	0.86
Toys & Hobbies	Single Hill	$-6,379 \pm 89$	36.4	57.3	0.94
	Mixture $K = 2$	$-6,346 \pm 89$	29.3	53.5	0.98
	Mixture $K = 3$	$-6,104 \pm 91$	34.3	47.3	0.96

Table 3: Real-data convergence pass rates (publication-ready criterion across three seeds), posterior mean effective K , the mean and maximum number of LOO observations with $\hat{k}_i > 0.7$ across seeds, and the posterior mean pairwise cosine separation between active Hill components (Section 7.4).

Organisation	Model	Pub. pass rate	$\mathbb{E}[\text{}^1D]$	$\hat{k}_i > 0.7$ (mean / max)	$\bar{d}_{\cos}^{\text{post}}$
Beauty & Fitness	Single Hill	0.67	1.00	1.7 / 3	—
	Mixture $K = 2$	0.00	1.65	15.0 / 21	0.46
	Mixture $K = 3$	0.00	2.17	3.3 / 10	0.68
Home & Garden	Single Hill	0.67	1.00	0.7 / 1	—
	Mixture $K = 2$	0.33	1.94	0.7 / 1	0.09
	Mixture $K = 3$	0.00	2.00	1.7 / 3	0.12
Toys & Hobbies	Single Hill	1.00	1.00	2.7 / 4	—
	Mixture $K = 2$	0.67	2.00	1.7 / 2	0.47
	Mixture $K = 3$	0.00	2.45	3.3 / 4	0.29

The combined reading is that the effective- $K \approx 2$ ceiling reported in Table 3 is not a generic property of real e-commerce MMM data, but an organisation-dependent one. Home & Garden lives at the resolvability boundary; Toys & Hobbies actually carries three resolvable components; Beauty & Fitness has strong heterogeneity but poor convergence. The synthetic resolvability framework therefore provides a concrete vocabulary for diagnosing why mixture-of-Hill behaves the way it does on a specific organisation, rather than treating all real data as a single black box.

8 Discussion

Mixture helps, but mainly in predictive density. Across both synthetic and real data, the most consistent benefit of mixture-of-Hill over single-Hill is in ELPD-LOO. The CRPS gains are smaller, and improvements in MAPE are organisation-specific. This pattern—predictive density improves first—is consistent with the view that mixtures primarily improve probabilistic calibration

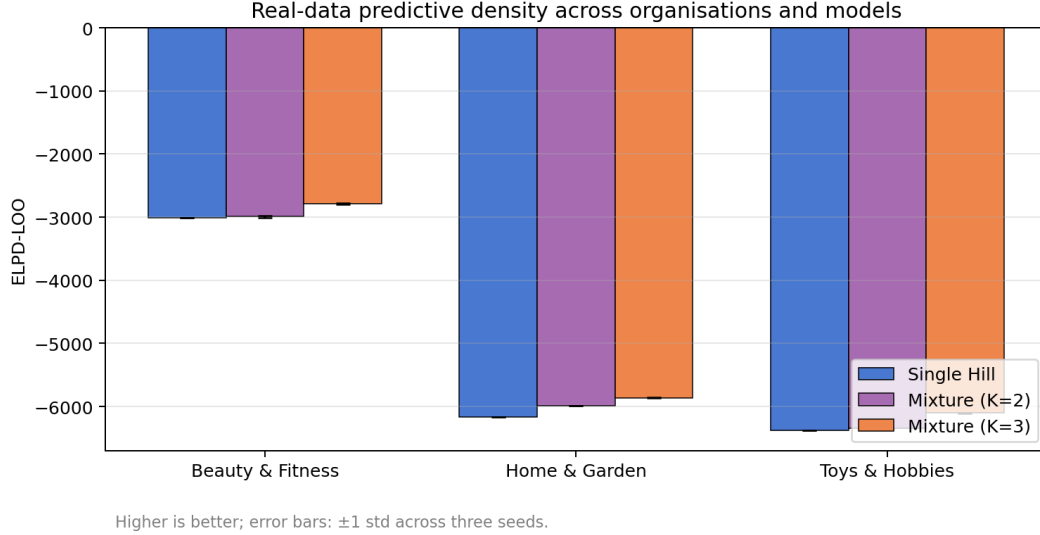


Figure 7: Real-data ELPD-LOO across the three organisations and three models. Higher is better. Error bars are ± 1 standard deviation across three random seeds. Mixture $K = 3$ achieves the highest ELPD-LOO in every organisation, with seed-level variability much smaller than the model-level gap.

and uncertainty representation rather than mean prediction. A practitioner whose objective is point forecasting may gain little; a practitioner who needs calibrated predictive intervals for budget decisions gains more.

The synthetic–real diagnostic gap. The publication-ready convergence criterion that passes on 100% of synthetic fits fails on 0/9 Mixture $K = 3$ fits on real data. This is not a contradiction: synthetic data are generated within the model class, with controlled noise and stationary mixing weights. Real organisations exhibit additional structure—seasonality, regime shifts, channel-mix changes, and outliers—that the within-model strict criterion is sensitive to. We report this gap rather than relaxing the criterion, because the gap itself is one of the most informative findings: it tells practitioners that synthetic-benchmark convergence does not automatically transfer.

Effective K is organisation-dependent. The headline two-effective-components figure in Table 3 hides organisation-level variation that Section 7.4 makes explicit. Home & Garden sits on the resolvability boundary in posterior cosine separation and the data genuinely cannot support more than ≈ 1.7 effective components. Toys & Hobbies under Mixture $K = 3$ has posterior separation 0.29 and Shannon count 2.72, which is inside the resolvable region: a three-component story is data-supported even though our strict diagnostic criterion still flags the fit. Beauty & Fitness shows the strongest mixture/single-Hill ELPD gap, the highest posterior separation, and the worst publication-readiness rate. The right read of these three cases is not “mixture saturates at two” but rather “mixture-of-Hill has organisation-specific identifiability and the resolvability framework lets us see which case each organisation falls into”.

Beauty & Fitness as a cautionary case. The Beauty & Fitness organisation has the strongest mixture-versus-single ELPD gap but also the most striking calibration failure: 90% holdout coverage rises from 0.33 to 0.54 across models but never approaches the nominal level. The high Pareto- k

Posterior response curves — Home & Garden (seed 0)

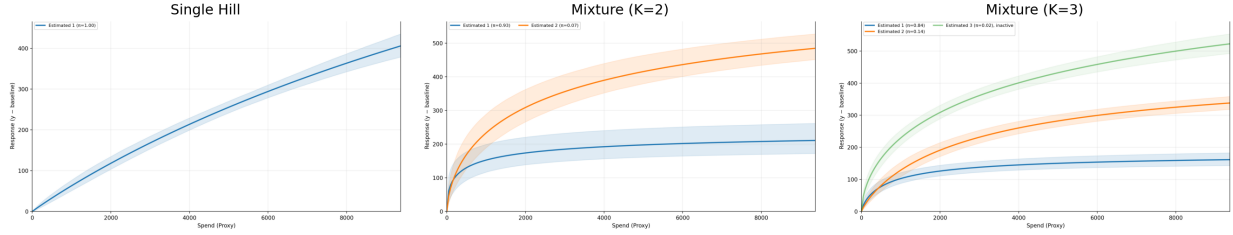


Figure 8: Posterior response curves for Home & Garden (seed 0) under Single Hill, Mixture $K = 2$, and Mixture $K = 3$. Shaded bands are 90% posterior intervals; legend entries report posterior mixture weights $\hat{\pi}$ per component. The Mixture $K = 3$ third component is labelled *inactive* because its weight collapses to 0.02.

count further indicates influential observations. The honest reading is that the mixture is doing real work but is still not a sufficient model for this organisation, and that a heavier-tailed likelihood or richer observation model would be needed before deploying it for decisions.

Identifiability caveat. As Dew et al. [2024] note, nonlinear and time-varying effects in MMM are jointly under-identified from typical observational time series. Our mixture-of-Hill model does not resolve this concern; it offers a controlled extension of the standard family in which the source of nonlinearity (heterogeneity, in this case) is at least made explicit. Stronger causal claims would require external variation, such as geo-experiments or incrementality tests.

9 Limitations

We separate limitations from open conclusions so that each can be addressed independently in follow-up work.

Data and scope. The real-data evaluation covers three organisations from three verticals of one e-commerce dataset; cross-organisation generalisation claims would require a substantially larger panel. We aggregate spend across channels into a single x_t in both benchmarks, which simplifies inference but loses cross-channel information. Channel-level mixture-of-Hill is a direct extension, with the caveat that label switching now appears within *each* channel.

Model structure. The model assumes mixing weights that are constant over time. Several of the publication-ready failures on real data are consistent with regime shifts (seasonality, channel-mix changes) that this assumption cannot absorb. The Gaussian likelihood is also not robust to influential observations, as seen most clearly in the Beauty & Fitness Pareto- k counts; a Student- t or heavy-tailed alternative would be a natural robustification.

Diagnostics. The publication-ready criterion is strict by design ($\hat{R} \leq 1.01$, $\text{ESS} \geq 100$ per chain, zero divergences). Real-data fits routinely fail this bar even when posterior summaries look qualitatively reasonable. We have not yet investigated whether a softer, application-aware criterion would distinguish “failure caused by misspecification” from “failure caused by posterior multi-modality with otherwise-good summaries”.

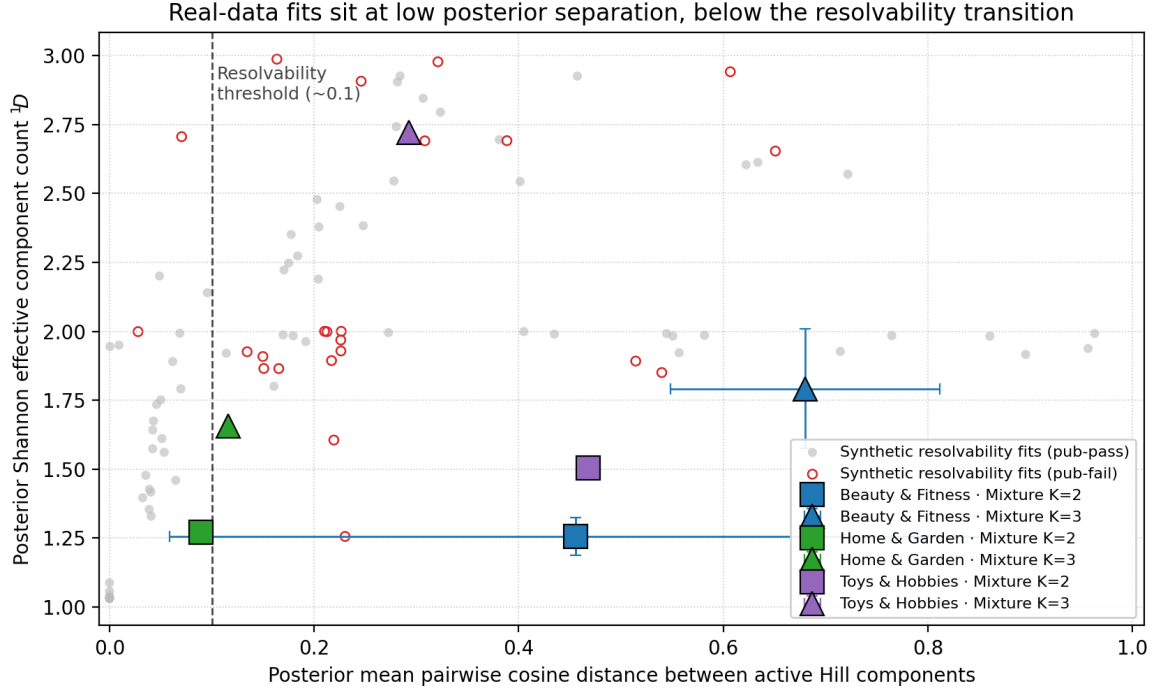


Figure 9: Posterior Shannon effective component count versus posterior mean pairwise cosine distance for the 90 synthetic resolvability fits (grey, with red-outlined fits failing the publication-ready criterion) and the three real organisations averaged across seeds (large coloured markers; error bars are ± 1 s.d. over seeds). The dashed line marks the cosine ≈ 0.10 resolvability transition identified in Section 6.

Priors and scales. The prior on λ depends on the empirical spend range, and the prior on A on $\text{range}(y)$. These data-dependent priors are operationally convenient but should be made explicit in any deployment. We have not run a sensitivity sweep on the multiplicative constants in those priors.

Causal identification. As above, the mixture extension does not address the joint identification problem in Dew et al. [2024]. Practitioners using this model for budget decisions should triangulate against geo-experiments or incrementality tests rather than relying on observational ELPD alone.

10 Code and Data Availability

All code, configurations, and per-case artifacts are released at <https://github.com/jpcca/marketing-mix-model>. The repository contains: model definitions and inference utilities (`src/hill_mixture_mmm/`), the synthetic data generator (`data.py`), the synthetic and real-data benchmark drivers (`scripts/run_benchmark.py`, `scripts/run_real_data_validation.py`), the resolvability sweep (`scripts/run_component_resolvability_sweep.py`), and the analysis script that computes posterior cosine separation for Section 7.4 (`scripts/compute_posterior_sep.py`). Per-fit JSON summaries containing posterior weights, label-invariant and relabeled diagnostics, and convergence flags are checked in under `paper/figures/{synthetic,real}/`. Random seeds are fixed and listed alongside the per-fit artifacts.

11 Conclusion

We presented a Bayesian mixture of Hill saturation functions for Marketing Mix Modeling, together with a label-invariant diagnostic workflow and three complementary benchmarks. Mixture-of-Hill consistently improves ELPD-LOO under heterogeneity, with mean improvements of +222 to +308 over single-Hill on the three real organisations; the gains appear primarily in predictive density rather than point prediction. The component resolvability study identifies an empirical cosine-separation threshold of ≈ 0.10 below which the posterior collapses to fewer effective components. Re-using this axis on the real fits diagnoses each organisation: Home & Garden lives at the resolvability boundary, Toys & Hobbies under Mixture $K = 3$ carries three genuinely resolvable components, and Beauty & Fitness has strong heterogeneity but a posterior whose chains do not mix sufficiently to certify it. The strict publication-ready convergence criterion that succeeds uniformly on synthetic data fails uniformly for Mixture $K = 3$ on real data; we interpret this gap as a directly useful finding for practitioners deploying mixture MMMs rather than a weakness to be sanded away.

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A Inference Hyperparameters

Table 4 reproduces the per-model NUTS settings used throughout the paper. Two chains are run in parallel for every fit; warmup uses NUTS dual averaging with a window size matched to the warmup length. Tree depth is capped at 10 and the initialisation strategy is `numpyro.infer.init_to_uniform`.

Table 4: Per-model NUTS configuration. Synthetic and real-data fits differ only in warmup/sampling length; all other settings are shared. The single-Hill synthetic schedule uses 600 warmup/samples on the $K = 1$ and $K = 2$ DGPs and 800 on the $K = 3$ DGP; mixtures use the same schedule across DGPs.

Model	Warmup (synthetic / real)	Samples (synthetic / real)	Chains	Target accept
Single Hill	600–800 / 600	600–800 / 600	2	0.90
Mixture $K = 2$	900 / 900	900 / 900	2	0.95
Mixture $K = 3$	1,600 / 1,600	1,200 / 1,200	2	0.97

B Priors

Table 5 lists the priors used in every experiment. The priors on A_k and μ_0 scale automatically with the training-data range and mean to accommodate diverse target magnitudes; the prior on λ uses a cumulative-sum reparameterization (Section 3.1) with positive log-normal increments. We have not run a formal sensitivity sweep on the multiplicative constants; that is noted as a limitation in Section 9.

Table 5: Priors. Statistics with hats (e.g. $\widehat{\text{range}}(y)$, $\bar{y}$, $\hat{\sigma}_y$) are computed from the training portion only.

Parameter	Distribution	Notes
Adstock decay α	Beta(2, 2)	shared across components
Half-saturation increments δ_j	LogNormal($\log(s_{\max}/(K+1))$, 0.7)	$\lambda_k = \lambda_1 + \sum_{j \leq k} \delta_j$
Hill slope n_k	LogNormal($\log 1.5$, 0.4)	per component
Component amplitude A_k	LogNormal($\log(0.3 \cdot \widehat{\text{range}}(y))$, 0.8)	per component
Mixture weights π	Dirichlet($\mathbf{1}_K$)	symmetric
Intercept μ_0	$\mathcal{N}(\bar{y}, 2\hat{\sigma}_y)$	data-dependent
Trend β	$\mathcal{N}(0, 0.5)$	on standardised t
Observation noise σ	HalfNormal($\hat{\sigma}_y$)	data-dependent

C Synthetic DGP Parameters

Table 6 summarises the parameter ranges used to generate synthetic data for the three DGPs in Section 5 and the nine profiles used in Section 6. Per-component values are drawn uniformly from

the listed ranges and held fixed across the five seeds within each profile.

Table 6: Synthetic DGP parameter ranges. The resolvability profiles fix A_k and n_k and vary only the spacing δ of the half-saturation points λ_k around a fixed centre c .

Parameter	Single ($K = 1$)	Mixture $K = 2$	Mixture $K = 3$
α	0.5	0.5	0.5
λ_k	[1.0, 2.5]	[0.8, 2.8]	[0.7, 3.0]
n_k	[1.0, 2.0]	[1.0, 2.0]	[1.0, 2.0]
A_k	[80, 120]	[80, 130]	[80, 140]
π_k	1	{0.6, 0.4}	{0.5, 0.3, 0.2}
σ	5.0	5.0	5.0
T	200	200	200
Train ratio	0.75	0.75	0.75